

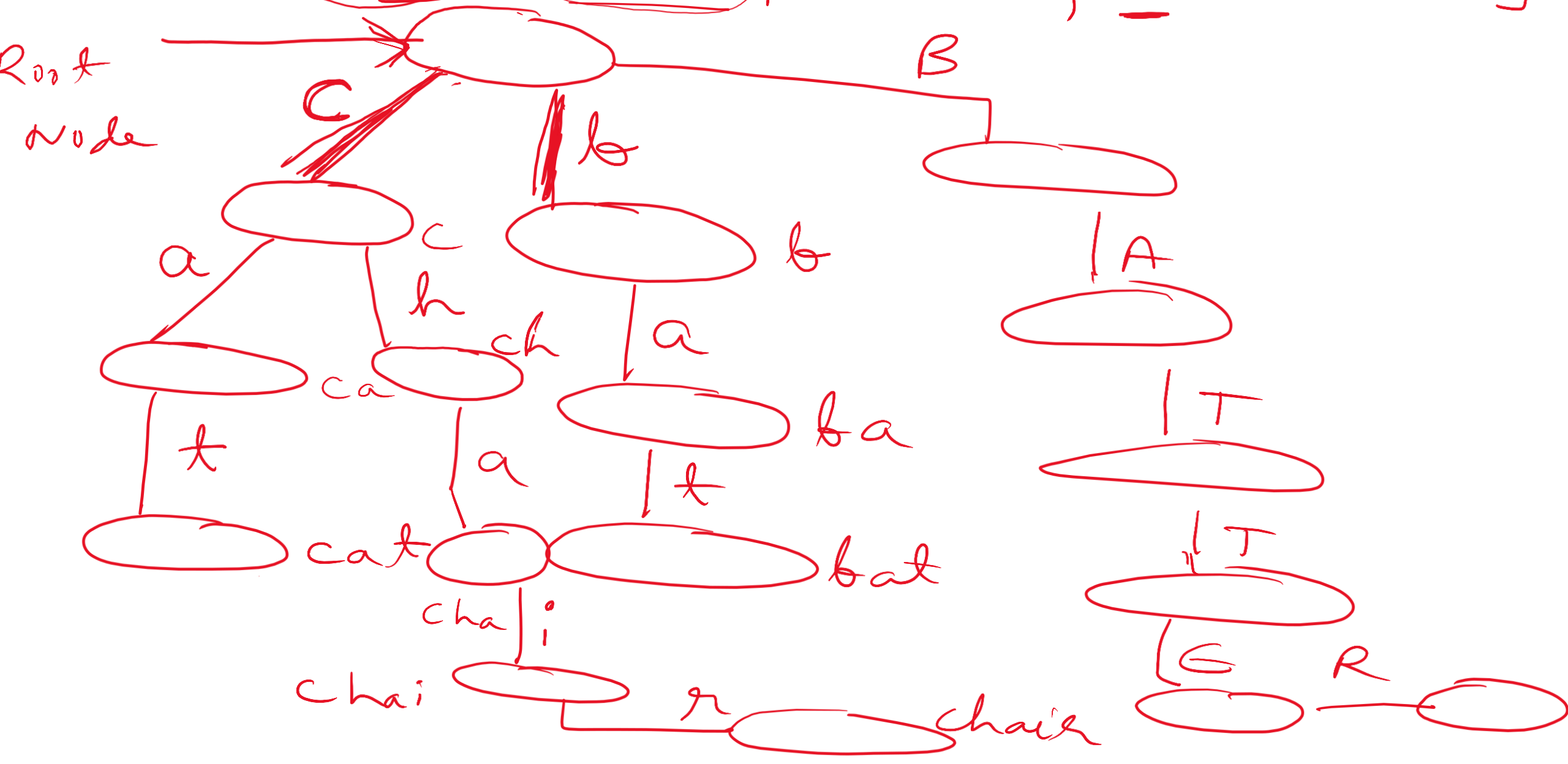
① Trie :- It is a tree type data structure which is used to store dynamic set of string or keys which are associated with text;

↳ Each Node in a trie data structure represents a single character of a string.

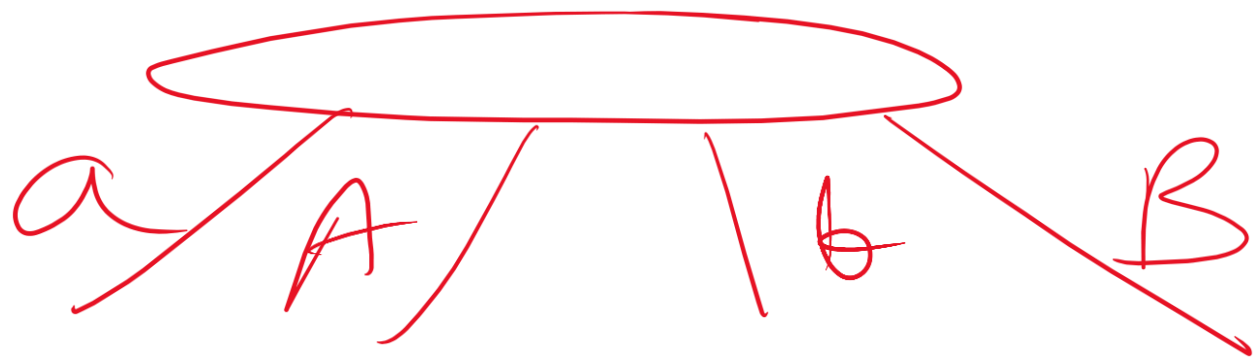
2. Root Node is Empty and Children Node corresponds to a character.

3. Path from Root Node to Leaf Node will form a String and Complete String will get formed only after leaf Node has been Reached.

eg:- [cat bat, chair, BATTER]



① In case of Trie Data Structure, interviewer will specify in the question that it consists of small letters or capital letters words or Both.



① Trie data structure is used when you need to perform searching operation and prefix matching operation.

② In any trie related problem, you need to perform insert operⁿ and then Search operⁿ.

① class TrieNode

{

TrieNode[] children = new TrieNode[26],

~~boolean~~ isEndOfWord;

TrieNode()

{ for (int i = 0; i < 26; i++)

{ children[i] = null;

}

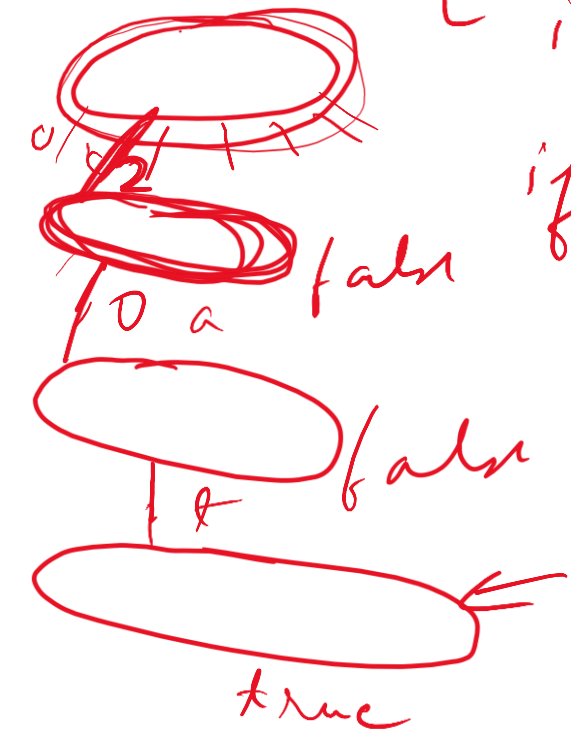
isEndOfWord = false;

}

```

① public static void insert (String word)
{
    TrieNode pcrawl = root;
    cat for (int i = 0; i < word.length; i++)
    {
        int index = word.charAt(i) - 'a';
        if (pcrawl.children[index] == null)
        {
            pcrawl.children[index] = new
                                   TrieNode();
        }
        pcrawl = pcrawl.children
                [index];
    }
    // for closed

```



pCrawl.isEnd of word = true;

// insert & closed.

```
① public static boolean search (String word)
{ TrieNode pcrawl = root;
  for (int i = 0; i < word.length; i++)
  { int index = word.charAt(i) - 'a';
    if (pcrawl.children[index] == null)
    { return false;
    }
    pcrawl = pcrawl.children[index];
  } // for loop closed
  return pcrawl.isEndOfWord;
}
```